

Measurability of the calibration entries

Direct statistical counterparts, indirect constructions, and latent model quantities

This revision assesses **all 35 grouped entries** in `V6_calibration_core.tex`, retaining their order and IDs. It follows AHP's distinction between constructing data and using a structural model to explain those data. Sources are the supplied AHP paper, the V6 equations and zero-exponent solver, existing empirical mappings, the 22 September API audit and a new foreign-dividend probe. Paper references use *journal page numbers*.

Separate measurement from assumptions used in calibration

Class	Meaning
D: directly measured	Reported statistics and transparent accounting constructions: sums, differences, ratios, unit conversions and revaluation-based indexes. No HKT equilibrium or latent parameter is needed to construct the specified empirical target. D does not mean error-free; the dagger flags population, instrument, timing or numeraire conditions for applying it to V6.
I: indirect / model implied	A model coordinate needs an unobserved input, a substantive proxy, a normalization or a structural inversion. A country-coverage approximation remains explicit. Ordinary subtraction or aggregation alone is not grounds for I, and a model-generated value is not independent evidence.
L: structurally latent	No verified observation rule measures the preference, technology or regime construct. It requires joint structural identification, a compatible external estimate, or an explicitly imposed restriction. L does not mean dispensable or inherently unidentifiable.
D/I: component split	A grouped entry contains both directly measured targets and unresolved model coordinates or country components. This is not a fourth degree of measurability: the row identifies which component is D and which remains I.

Assessment. Seven families are classified D, six I, seventeen L, and five D/I. In particular, **M34 equity valuation and M35 CA/NFA changes are D**; M28 compensation is also D under an explicit population convention. M22 portfolio weights, M27 skill wages, M29 prices/dividends, M30 corporate value/cash flow and M32 bonds are split by their observable and unresolved components. These family counts do not imply that every country component is already collected.

What AHP adds. Residence-based corporate GVA, labor compensation, enterprise value/FCF, external positions, revaluations, flows and foreign monetary-dividend yields provide concrete calibration targets. Corporate financing neutrality is a legitimate maintained interpretation, as in AHP. V6's new-variety issuance, skill composition and research technology require additional accounting or structural work; accurate measured aggregates do not identify those primitive parameters by themselves.

Verification. The earlier WRDS/FRED/Eurostat receipts are preserved. This revision reproduces the saved AHP external accounts, 36 annual observations each of US GVA, compensation, EV and FCF, and a 10-quarter AHP-style foreign cash-yield sample. It revises the measurement assessment without re-estimating the model. AHP's separate Supplemental Appendix and original author workbook were not available in the supplied reference.

1 Parameters: saving, portfolios and regime risk

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M01 β	L Structural identification	AHP calibrates its time-preference rate from an average dividend yield using its balanced-growth model. This is a structural restriction, not a measurement of V6 β . National net saving/DPI differs from cohort asset acquisition/labor income A/e_{US} ; select the OLG period and identify jointly.	AHP, S1 AHP Sec. III, p. 2175; BEA saving and SCF data are supporting evidence, not a direct β series.
M02 γ	L Structural identification	Risk aversion in the specified continuation utility has no data column. Risk-attitude answers and realized returns can discipline a structural preference estimate, but neither equals this curvature. External estimates require a compatible utility specification and population.	S1, F2 SCF questionnaire; CRSP returns via WRDS. Survey metadata and live return sample.
M03 κ	L Structural identification	Portfolio data discipline a return-wedge/target-deviation combination, not this preference curvature alone. A single equity share cannot separate κ from $\bar{\omega}$, risk preferences and expectations. Requires allocation responses or explicitly imposed restrictions.	F2, F3 CRSP returns; resident equity holdings. Neither source directly measures utility costs.
M04 $\bar{\omega}, \bar{\omega}^*$	L Preference centers	AHP measures ownership fractions; V6 $\bar{\omega}, \bar{\omega}^*$ are preference centers. Measured actual portfolio weights (M22) help discipline these centers jointly with return wedges and curvature. Equating a preference center to observed home bias imposes an extra restriction.	AHP, F3 AHP App. A4, pp. 2198–2199; source-backed ownership ratios cannot directly identify preference parameters.
M05 χ	L Structural identification	A safe-asset spread or foreign Treasury share does not equal convenience utility. The model links χ jointly to holdings, marginal valuation and the safe-rate wedge; it also changes RoW saving. Currency, credit and duration differences in observed bonds must be separated from convenience.	F6, S1 H.15/TB3MS and Treasury TIC holdings; household saving evidence. No exact χ series.
M06 η	L Structural identification	The coefficient scales a utility cost of the bond share, not observed underwriting fees or a resource cost. It needs joint identification from holdings and risk-adjusted return wedges. $\Upsilon(\theta) = -\log(1 - \theta) + \theta^2/2$ is imposed by the code.	F2, F6, M1 Equity returns, bond yields/positions; model first-order condition. Functional form is fixed.
M07 π	L Regime definition missing	Persistence becomes estimable only after an operational definition of u/b and a period length. A recession indicator or a chosen 2008 date is not the model technology regime. An all- u episode gives censored duration evidence, not a directly observed interior transition probability.	M1, P5 No verified regime series. Productivity/investment histories are possible inputs to a future state rule.

2 Parameters: production and knowledge

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M08 H_i, L_i	D[†] Conditional: scope and units	Direct employment counts/hours exist for declared skill groups. Candidate: US CPS BA+ versus exhaustive non-BA; Eurostat tertiary versus non-tertiary, age 25–64. Adopt a common age, sector, headcount/FTE and country convention. Education as model skill is a maintained bridge; efficiency units remain I.	P1 CPS ORG: local micro mapping. Eurostat 1f5a_egaed: live DE sample, annual thousand persons.
M09 a_i	L Structural identification	$a_i = (N_{i,t+1}/N_{i,t} - 1)/[(1 - \varphi_i)H_i]$ is conditional recovery, not direct measurement: neither knowledge varieties nor matched research labor is yet observed. Free-entry inversion uses the same valuation bridge. Separating a_i from $a_i H_i$ needs the labor unit and model period.	P2, P3, P1 MSTI/BERD research inputs; R&D/patent stocks; labor quantities. Existing implied- a series is a diagnostic.
M10 ϑ_i	L Structural identification	AHP infers its output wedge from valuations, capital and discounting; it does not directly observe V6 inverse markup ϑ_i . V6 recovery from profits and skilled production wages also needs φ_i , sector alignment and treatment of taxes/capital. Profit shares are useful targets; the markup remains structurally identified.	AHP, F1, P4 AHP Sec. III, pp. 2175–2176; IMA earnings/compensation and Compustat supply evidence.
M11 α_i	L Normalization / identification	AHP infers a Cobb–Douglas cost share conditional on its output wedge. V6 uses a different CES weight: levels depend jointly on $\alpha_i A_{X,i}^{1-\rho_i}$ and $(1 - \alpha_i) A_{L,i}^{1-\rho_i}$. Observed labor shares do not separately identify that weight and augmentation scales.	AHP, P1, P4 AHP Sec. III, p. 2176; measured labor shares are targets, not the V6 coefficient.
M12 ρ_i	L Structural identification	The elasticity is $1/\rho_i$. Identification needs relative-input/price variation and assumptions on markup and augmentation dynamics. A generic capital–labor elasticity is not automatically the intermediate-input/unskilled-labor elasticity used here. Both countries use CES in the solver.	P1, P4, M1 Labor quantities/prices; production accounts; compatible external structural estimates.
M13 $\bar{A}_{X,US,u}, \bar{A}_{L,US,u}$	L Structural identification	Factor-specific scale coefficients need a production inversion conditional on α, ρ, ϑ , knowledge normalization and output/input units. R&D spending, sales per employee, and industry TFP do not measure these coefficients.	P1, P4, P5, F1 Labor/production accounts and firm R&D are ingredients. No verified direct augmentation series.
M14 $\bar{A}_{X,US,b}, \bar{A}_{L,US,b}$	L Counterfactual restriction	Require measured absorbing-regime production with a justified state rule, or a counterfactual calibration restriction. An all- u sample does not reveal absorbing productivity. Labeling a historical crash as b does not establish the technology correspondence.	P4, M1 Production histories plus an independently specified regime rule; currently no exact branch mapping.
M15 $\bar{A}_{X,W}, \bar{A}_{L,W}$	L Structural identification	RoW augmentation scales need harmonized quantities, wages and output under the solver CES. A sector productivity ratio or a weighted country average is not either augmentation. Covered G4/OECD-ex-US panels are not the entire residual world.	P1, P4, P5 Eurostat/OECD labor and institutional accounts. Matched RoW wage levels and scope remain gaps.
M16 ξ_u, ν_u	L Structural identification	Elasticities of separately inferred augmentations with respect to a separately mapped knowledge stock. Relative wages alone cannot identify both. Defining $A_X = N^{\xi_u}$ and comparing that generated series to N imposes the elasticity rather than measuring it.	P3, P1, P5 Knowledge proxies and production evidence; no directly measured elasticity pair.
M17 $\nu_b = 0, \xi_W = 0$	L Imposed, not estimated	Known by construction in the chosen solver; not measurements of zero elasticities. Retain as maintained restrictions and test implications. Positive R&D investment growth does not by itself reject zero factor augmentation. Reopening these coefficients changes the computational specification.	M1, P5 <code>validate_params</code> and zero-exponent restriction. R&D investment is only contextual evidence.

3 Exogenous inputs

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M18 $N_{US,0}, N_{W,0}$	I Normalization / proxy	Setting initial knowledge to one is a normalization, not a measured level. A patent or R&D-capital proxy requires office/residence, timing, depreciation and initial-stock conventions. Rebase only after these are fixed; normalized units do not remove the proxy gap.	P3 Fed/BEA R&D stocks; USPTO/OECD/WIPO patent alternatives in the local package.
M19 $z_t \in \{u, b\}$	L Empirical state rule missing	Exogenous in the model does not mean observed in data. A switch date τ imposed in a simulation is a scenario. A documented empirical regime classifier would yield an inferred state (I); no accurate operational u/b mapping is presently supplied.	M1 Code path available; no verified empirical state series.

4 Endogenous states and equilibrium choices

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M20 $N_{i,t}$	I Model state / proxy	The law of motion gives a model-implied path. R&D-cost stocks and perpetual-inventory patent stocks are observable constructions but do not count model varieties. Their growth depends on valuation, depreciation, office coverage and initialization; preserve distinct proxy families.	P3 Q4 R&D stocks FL105013465, FL795013465; patent stock alternatives. Local evidence.
M21 $\varphi_{US,t}, \varphi_{W,t}$	I Research/skill bridge missing	Need $\varphi = 1 - R/H$ with research hours/persons a subset of the same skilled population. $NRC \cap H/H$ fixes the denominator but remains a task proxy: management/clinical work is not necessarily innovation. BERD/MSTI researcher or personnel FTE and headcount are different definitions.	P1, P2 CPS/NRC and BERD/MSTI. Live OpenEcon researcher response rejected for wrong measure; OECD HTTP 403 retained.
M22 ω_t, ω_t^*	D/I D: US weight; I: RoW scale gap	Given an adopted operating-asset/equity perimeter, $\omega = (V_{US} - E_L)/(V_{US} - E_L + E_A)$ is an accounting construction. $\omega^* = E_L/(V_W - E_A + E_L)$ additionally needs the full RoW value. AHP observes E_A without measuring V_W directly. Its foreign ownership ratio E_L/V_{US} is neither ω^* nor $\bar{\omega}^*$.	AHP, F3, F7 AHP App. A4, p. 2198; US EV and gross equity positions exist locally. Full-world denominator remains a gap.
M23 $\theta_{US,t}^*$; recovered $\theta_t, \theta_{W,t}^*$	I Foreign wealth / safe-claim bridge	AHP net non-equity accounts can anchor the numerator, but V6 foreign bond share needs matched RoW investable wealth. AHP App. A4 does not supply that denominator. Broad non-equity wealth is not automatically safe. Recovered US shares can be constructed after choosing that aggregation; $\theta_W^* = 0$ remains a model restriction.	AHP, F4, F6 AHP Eq. (25) and Sec. III footnote 23; TIC holdings alone do not measure the portfolio denominator.
M24 $R_{f,t}, R_{f,t}^W$	I Real payoff / shadow yield	AHP recovers its cost of capital through investment and current-account equations; this is model inference, not a directly observed V6 safe rate. A named bill yield is observable, but needs maturity, discount-quote and numeraire conversion. Ex-post deflation is not an ex-ante safe real payoff; the zero-supply RoW yield remains model implied.	AHP, F6 AHP Sec. III, pp. 2175–2176; H.15/TB3MS observed through Aug 2026 in the saved probe.

5 Endogenous production and valuation

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M25 $A_{X,i,t}, A_{L,i,t}$	I Conditional production inversion	AHP sets aggregate productivity growth to real corporate-output growth within its model. That rule does not measure V6 factor-specific A_X, A_L . Recover these only conditional on CES parameters and matched quantities/wages. Corporate GVA and labor shares can discipline the block without separately observing augmentations.	AHP, P1, P4 AHP Sec. III, p. 2175; IMA/OECD output and compensation are usable inputs, not augmentation series.
M26 $Y_{i,t}$	D[†] Measured output; scope condition	Use the AHP residence-based corporate GVA aggregate: IMA S.5+S.6, with nonfinancial-only sensitivity; OECD S11+S12 for covered countries. GDP is a distinct denominator. Reconcile capitalized R&D with V6 $Y = e + \mathcal{D} - \mathcal{I}$. AHP estimates Canadian/Japanese corporate GVA through a share proxy; those country estimates remain I.	AHP, P4, P5 AHP App. A2/A5, pp. 2194–2200. Saved US GVA construction reproduced for 36 annual observations.
M27 $w_{H,i,t}, w_{L,i,t}$	D/I D: US earnings; I: RoW levels gap	US mean earnings per compatible labor unit and skill group are direct candidates. Earnings exclude benefits; median weekly wages are not mean hourly compensation. Corporate-compensation scaling is I. Current RoW relative-earnings panels do not supply matched wage levels, so that component remains I/gap.	P1, P4 US CPS ORG micro wages; BLS wage sensitivities. OECD EAG ratios and Eurostat labor quantities: local mappings.
M28 $e_{i,t}$	D[†] Measured compensation; population condition	AHP directly constructs corporate labor compensation. The same IMA/OECD employee-compensation aggregate is a usable counterpart to $e = w_H H + w_L L$ after adopting a common population and labor-income boundary. It need not be inferred from equilibrium or first reconstructed from skill wages. This does not separately measure skill wages or OLG saving propensity.	AHP, P4, P1 IMA FU106025005+FU796025005; OECD D1, S11+S12. Saved US compensation reproduced for 36 years.
M29 $q_{i,t}, d_{i,t}$; reconstructed $R_{i,t+1}$	D/I D: price/yield moments; I: levels	AHP Eq. (3) gives observed revaluation returns G_t/E_{t-1} and a chained price index. App. A4 constructs foreign cash-dividend yield from receipts divided by lagged holdings plus revaluation. These are direct accounting moments with scope/FX qualifications. Absolute per-variety q, d still need N ; total return needs matched payouts.	AHP, F3, F5, F8 App. A4, pp. 2198–2199. New exact NIPA dividend probe supports 10 quarterly yield observations; see timing formulas.
M30 $\mathcal{Q}_{i,t}, \mathcal{D}_{i,t}$	D/I D: EV/FCF targets; I: payout bridge	AHP enterprise value and FCF are directly constructed and legitimate targets under its fully equity-financed operating-asset convention. This can guide V6 $\mathcal{Q}, \mathcal{D}$. However, V6 pays incumbent profits while IPO proceeds fund new varieties. Reconcile net investment/issuance before equating AHP FCF to $\mathcal{D}$; retain equity-capitalization and earnings alternatives.	AHP, F7, P4 AHP Sec. I, Eq. (13), App. A2–A5. US EV/FCF replicated locally; RoW country panels have coverage/sign qualifications.

6 External positions, valuation and flows

Core entry	Class / qualifications	Mapping and calibration use	Data source / verification
M31 $qW,t n_{W,t}, q_{US,t} n_{US,t}^*$	D[†] Measured external equity positions	AHP observes US equity assets abroad E_A and foreign equity in the US E_L directly from IMA positions. V6 can use these aggregate values without observing claim counts or the full RoW equity market. Match residence, portfolio/FDI/fund scope, timing and currency. Claim counts remain inferred only if needed separately.	AHP, F3 App. A1: historical S.9 lines 153/125. Exact FRED ROWEQ027S / ROWEQ027S already saved.
M32 $b_t = \theta_t A_t$	D/I D: net non-equity; I: safe-bond interpretation	AHP constructs $b^{NE} = NFA - E_A + E_L$ directly from positions. The CA-closing non-equity flow $F^{NE} = CA - (P_A - P_L)$ is also observable arithmetic. It need not equal Δb^{NE} because valuation/other changes intervene. Using either as V6 b or Δb requires a stated single-bond aggregation; no latent parameter is needed for the empirical residual.	AHP, F3, F4 App. A1 and Eq. (25); both residual constructions are available in AHP_NFA. Preserve stock-versus-flow distinction.
M33 NFA_t	D[†] Measured NFA / changes	US NFA is external assets minus liabilities, or negative ROW net worth. This is an observed accounting target and is used in Notebook 06. Sector/instrument omissions concern model fit and accounting correspondence, not whether NFA can be measured. Keep the chosen empirical perimeter and GVA denominator fixed.	AHP, F4 App. A1; ROWTLEQ027S-ROWTASQ027S = -ROWNETQ027S. Saved observations and period targets independently rechecked.
M34 $VA_t^{asset}, VA_t^{liability}, VA_t$	D[†] Measured equity revaluations	AHP takes gross equity revaluations from IMA and sums them with the US-liability minus sign. V6 can target $VA^{asset}, VA^{liability}, VA^{eq}$ without separately identifying n and q . FX, fund coverage and non-equity valuation belong in the mapping conditions; observed valuation is not an independently measured bubble component.	AHP, F5, F3 App. A1: historical S.9 lines 100/84; FR263181105 and negative FR263081115 in the saved notebook. Broad-fund alternative retained.
M35 $CA_t, \Delta NFA_t$	D[†] Measured CA and NFA change	AHP defines $CA = -RWBACQ027S/4$ and differences observed NFA. These are direct accounting targets, as in Notebook 06. Retain total/non-equity valuation and the residual in empirical closure. The model identity $\Delta NFA = CA + VA^{eq}$ is a restriction to assess, not a reason to classify the data as indirectly measured.	AHP, F4, F5 App. A1, p. 2193; exact saved 2008Q1–2023Q3 target reconstruction. Corporate-GVA normalization matches each notebook vintage.

7 Borrowing AHP's corporate measurement

AHP first constructs accounting quantities, then selects parameter paths to reproduce them (Sec. III, pp. 2174–2176). The second step does not turn the first-step observations into latent variables. Equally, the availability of those observations does not make the inferred preference or technology parameters directly measured.

V6 entry	AHP construction	How it can be used here
M26: output	Corporate GVA: historical IMA S.5+S.6; OECD S11+S12. Residence, rather than multinational headquarters, defines the economy.	Direct output/denominator candidate. Keep NIPA corporate GVA and the broader IMA financial-business aggregate as distinct alternatives. R&D capitalization must agree with the V6 production boundary.
M28: labor income	Compensation paid by the same corporate sectors, including employer contributions.	Direct aggregate-income candidate. Skill-specific wages and labor endowments are additional measurements; aggregate compensation need not wait for that decomposition. Literal age/cohort coverage remains a maintained bridge.
M30: corporate value	AHP V is the market value of operating/nonfinancial assets: equity plus other financing claims, less financial assets, with FDI/residence adjustments.	A legitimate candidate for an all-equity operating sector, rather than an invalid proxy merely because real firms issue debt. Retain shareholder-capitalization alternatives and use consistent claims in portfolio denominators.
M30: corporate cash flow	$FCF = NOS - TAX - NCF = GOS - TAX - GCF$; AHP interprets this as dividends of fully equity-financed firms with no financial assets.	Directly constructed net-cash-flow target. The exact V6 payout correspondence needs an investment/issuance decision, since incumbent dividends and new-variety finance are separate in V6.
RoW counterparts	AHP App. A5 uses OECD country accounts, with market-exchange-rate aggregation and a PPP sensitivity. Canadian/Japanese GVA is estimated using a corporate share proxy.	Covered-country panels support output, compensation, EV/FCF ratios and robustness comparisons. They do not constitute a directly measured full-world equity denominator. Preserve the OECD net-financial-worth sign as reported; any sign reversal needs its own mapping ID and balance-sheet check.

The substantive V6 cash-flow qualification. AHP's corporate financing neutrality makes EV a defensible valuation target (Sec. I, pp. 2162–2164; footnotes 16, 18). V6 also has no corporate debt, but its date- t entry financing obeys

$$Q_t = N_{t+1}q_t, \quad \mathcal{D}_t = N_t d_t, \quad \mathcal{I}_t = (N_{t+1} - N_t)q_t, \quad Y_t = e_t + \mathcal{D}_t - \mathcal{I}_t.$$

Under the *illustrative* national-accounts bridge $GVA = Y + \mathcal{I}$, compensation = e , gross investment = $\mathcal{I}$, and no taxes or other investment, AHP-style FCF is $\mathcal{D} - \mathcal{I}$, not $\mathcal{D}$. Thus the corresponding V6 payout would be $FCF + \mathcal{I}$. This is a conditional accounting derivation, not an empirical identity already established. Do not add capitalized R&D to observed GVA twice or mechanically add measured R&D spending to FCF without the required expenditure/issuance bridge.

What does not transfer as direct measurement. AHP's output wedge, cost-of-capital and production-share identification uses its own equations. Its Q is an investment-goods price; its H is human wealth; its ρ is a time-preference rate. These are not V6 Q , skilled labor H , or CES curvature ρ . AHP supplies useful moments but no direct observations of V6 innovation productivity, variety stocks, factor augmentations, skill-specific research shares or preference centers.

8 AHP ownership, prices and income yields

Observed ownership is different from an investor's portfolio weight. Write E_A for US external equity assets and E_L for US external equity liabilities. AHP App. A4 measures issuer ownership $1 - \lambda = E_L/V_{US}$. Under an adopted operating-asset/equity universe, V6 investor portfolio weights instead satisfy

$$\omega = \frac{V_{US} - E_L}{V_{US} - E_L + E_A}, \quad \omega^* = \frac{E_L}{V_W - E_A + E_L}.$$

The US ratio can be constructed from measured aggregates under the same perimeter. The RoW ratio needs V_W . AHP explicitly does *not* directly measure full RoW enterprise value in App. A4; it uses US foreign holdings and their payout yield. Accordingly, neither its ownership share nor an EU/G6 approximation automatically supplies V6 ω^* , and neither identifies a preference center.

AHP's foreign-yield route avoids the full-world denominator. Let G_t^A be positive-signed revaluation of US-held foreign equity and C_t^A monetary dividend receipts on that same basket. AHP Eq. (3) and App. A4 motivate the observable constructions

$$g_t^{q,A} = \frac{G_t^A}{E_{A,t-1}}, \quad P_t^A = P_{t-1}^A(1 + g_t^{q,A}), \quad y_t^{cash,A} = \frac{C_t^A}{E_{A,t-1} + G_t^A}.$$

Under matched holdings, payouts and the V6 homogeneous-claim interpretation, the last ratio corresponds to $d_{W,t}/q_{W,t}$. A corresponding cash total-return construction is $1 + (G_t^A + C_t^A)/E_{A,t-1}$, subject to within-period timing and instrument coverage. Absolute per-variety prices and dividends remain unobserved. Moreover, $\mathcal{D}_{i,t}/\mathcal{Q}_{i,t} = (N_{i,t}/N_{i,t+1})(d_{i,t}/q_{i,t})$: the per-variety yield is not the aggregate yield when new varieties enter.

New source check. NIPA Table 4.1 line 11 is FRED B3375C1Q027SBEA; quarterly monetary receipts in million USD equal the published billion-USD SAAR value times 1000/4. The exact-source probe gives ten usable quarters, 2024Q1–2026Q2. The 2026Q2 AHP-style cash yield is 2.487% at a simple annual rate (0.622% for the quarter). This is a monetary-yield candidate, not a measure of reinvested earnings or a guarantee of matched V6 payouts. Tax-driven repatriation spikes and currency/instrument scope remain explicit.

The external-account targets are measured. App. A1 retains

$$\Delta NFA = CA + VA^{eq} + VA^{non-eq} + RES.$$

The saved 2008Q1–2023Q3 moments, divided by endpoint corporate GVA, reproduce Notebook 06:

$$-1.023669 = (-0.521721) + (-0.671568) + 0.087168 + 0.082452.$$

The terms are NFA change, CA, equity VA, non-equity VA and the residual, respectively. Residual closure is algebraic, not an independent model-fit test. The liability price-index construction still needs the documented fund-coverage reconciliation between the broad stock and narrower revaluation series; this affects the precise index choice, not the observability of the reported revaluation itself.

9 Preserved API audit and revision evidence

Route / sample	Verified result and interpretation
WRDS: Compustat NA	<code>comp.funda</code> , <code>gvkey</code> 001690/ <code>fyear</code> 2024 returned 2 formats. Exactly 1 row is C/INDL/STD; the other is SUMM_STD. R&D, employee, payout and cash-flow fields exist in the standard row. This confirms a firm-level route, not national research/skill quantities. Duplicate-format summation would be wrong.
WRDS: Compustat Global	<code>comp.g_funda</code> , <code>fyear</code> 2024 returned an intentionally capped 3-row pilot; 42,376 matching rows were advertised. The pilot is incomplete and not representative. No complete Global panel, growth trend or aggregate moment was inferred.
WRDS: CRSP	<code>crsp.msf</code> , PERMNO 14593 returned all 12 unique months of 2024 with price, distribution-inclusive/exclusive return and adjustment fields. The provider count reported 3 despite 12 returned records. Calendar/identifier validation established the bounded sample; the inconsistent metadata were preserved. This does not certify coverage of all securities or a current national index.
Exact FRED / Fed data	19 exact series: 16 quarterly through 2026Q2, TB3MS through Aug 2026 and 2 annual dividend series through 2025. The request window began in 2024. Dates are unique/sorted; the NFA identity holds exactly in all 10 sampled quarters. These are saved-sample dates, not a claim about full source histories.
Exact Eurostat: labor	<code>lfsa_egaed</code> , DE, age 25–64, both sexes, ED5-8, THS_PER: 12,543.1; 12,708.1; 13,052.0 thousand employed persons in 2021–23. The 2021 break flag is preserved. This directly measures tertiary employment, not corporate skill efficiency or research labor.
Exact Eurostat: R&D	<code>nama_10_a64_p5</code> , DE, TOTAL, N1171G, P51G, CP_MNAC: 105,966; 110,894; 124,513 million national currency in 2021–23. The 2022–23 provisional flags are preserved. This is an investment flow, not knowledge varieties, innovation success or A_X .
OpenEcon: 22 September	Requested exact ROWEISQ027S, received FL263164100Q: the 2026Q2 values differ by 11,305,117 million USD. Requested MSTI business researchers B_RS/FTE, received G+T_RS URLs and generic units. Both responses contained numbers; both were rejected for these mappings. Direct exact-source probes supplied the usable evidence.
Failed routes retained	The exact OECD MSTI direct request returned HTTP 403; earlier saved OECD data remain separately labeled. Initial network/time-out routes and unsuccessful candidate endpoints are logged. Failure of a route is not evidence that the economic series does not exist.

Reproducible package. `measurability_evidence/` contains immutable-for-this-audit raw samples, request/response receipts, source IDs/URLs, flags, validation summaries and the classification CSV/JSON. WRDS metadata are in `wrds/tool_receipts.json`; official probes and rerun scripts are under `production/` and `external/`. The AHP revision adds three exact-source CSVs, a ten-quarter monetary-yield probe, and offline verification of saved corporate/external mappings under `ahp_revision_2026-09-23/`. Its OpenEcon request timed out; the receipt is retained. The PDF is rebuilt by `build_tables.py` plus LaTeX. A rerun can receive revised source data; retain this dated snapshot when changing vintages. No model solve, parameter estimation, bulk historical refresh or change to the earlier mappings was performed.

10 Data-source and verification ledger

Source keys used in the tables are defined below. Live sample dates identify the data actually retrieved; local panels retain their earlier vintage. URLs point to the producer or authenticated data endpoint. The accompanying source ledger records the full descriptions.

AHP. Atkeson, Heathcote and Perri (2025)

User-supplied published-paper transcription reviewed, 23 September 2026.

The End of Privilege, AER 115(7), 2151–2206. Sec. I and Eqs. (1)–(3): measured external accounts. Sec. III, pp. 2174–2176: data construction followed by model identification. App. A1–A5, pp. 2192–2200: exact measurement conventions. Local reference:

`Codes/Reference/Atkeson_2025_End_of_Privilege.tex`. Journal page numbers, not recompiled pages, are used here. Separate Supplemental Appendix and author workbook were not obtained; present data retain their own vintages.

Published article; Author data repository.

S1. Household saving and preferences

Official metadata checked; no new SCF microdata.

BEA defines personal saving relative to disposable personal income; SCF collects family assets, debts, income and attitudes. These observations do not directly identify the model utility coefficients.

BEA saving definition; SCF public data; SCF questionnaire.

P1. Skill labor and wages

Local CPS/EAG panels; live Eurostat sample and metadata.

US CPS ORG local baseline: private-for-profit employees, 1994–2024, BA+ versus exhaustive non-BA, headcount or matched-hours FTE. Eurostat `lfsa_egaed`: annual thousand persons, sex T, age Y25-64, ED5-8/ED0-2/ED3_4. Local RoW earnings ratios do not establish wage levels. BLS definitions; LFS metadata; Exact DE API sample.

P2. Research labor

Local official data/primary metadata; live OECD route 403.

OECD MSTI v1.3: business researchers B_RS or total personnel B_TT, FTE versus persons. NCSES BERD 2023 Tables 50/52 cover domestic business employment/research personnel; current local R&D FTE panel is 2015–2023. Survey scope and numerator membership in H must match.

OECD MSTI; BERD Table 52.

P3. Knowledge proxies

Existing frozen mappings; no exact-variety observation.

Fed/BEA Q4 R&D net stocks FL105013465+FL795013465; investment-price deflator Y006RG3Q086SBEA is a sensitivity, not an official stock-volume index. USPTO grants, OECD triadic and WIPO applicant-residence stocks have distinct coverage, initialization and depreciation.

NFC R&D stock; Financial R&D stock.

P4. Corporate output, labor income and equity

Existing 2026-09 mapping vintage; producer definitions checked.

US GVA: FU106902501Q+FU796902505Q; compensation: FU106025005Q+FU796025005Q.

OECD institutional accounts: B1G and D1 for S11+S12; F51 liabilities are a distinct equity alternative. Native currencies, market FX/PPP and end-period/flow timing are separate choices.

US corporate GVA component; OECD accounts.

P5. R&D investment and accounting

Live Eurostat data; BEA primary definitions.

Eurostat `nama_10_a64_p5`: N1171G asset, P51G transaction, TOTAL industry; CP_MNAC is current-price million national currency. BEA capitalizes R&D; expenditure and GDP contributions are not innovation success or factor augmentation.

Exact DE R&D API sample; BEA capitalization.

F1. WRDS / Compustat firm accounts

Live OPTIONS and data, 2026-09-22.

`comp.funda`: gvkey 001690/fyear 2024, 2 raw format variants, 1 C/INDL/STD row after local selection. `comp.g.funda`: intentionally capped 3-row 2024 pilot, not a complete or representative sample. Employee/R&D/accounting fields do not give research/skill hours or model markups. API types alone do not certify units.

NA API; Global API; Producer/access overview.

F2. WRDS / CRSP security prices and returns

Live 12-month 2024 sample; metadata discrepancy retained.

`crsp.msrf`, PERMNO 14593: `prc,ret,retx,shrout,cfacpr,cfacshr`. All 12 unique 2024 months returned. Source count said 3 despite 12 records; completeness was checked against the requested calendar. Security returns are observable, but do not identify a model variety or a national investor portfolio.

Monthly stock API; CRSP product description.

F3. International equity positions

Live exact FRED samples and Fed component definitions.

US external equity assets: ROWEISQ027S/FL263181105; US equity liabilities: ROWEINQ027S/FL263081005. Quarterly end-period million USD. Current fund/FDI/international-organization coverage must be read from the component trees, not inferred from an old shorthand label.

Asset component tree; Liability component tree.

F4. NFA, transactions and current account

Live samples 2024Q1–2026Q2.

US NFA=ROWTLEQ027S-ROWTASQ027S=-ROWNETQ027S, million USD. AHP App. A1 defines CA as negative ROW net lending/borrowing: -RWLBACQ027S/4. Retain this declared accounting convention and distinguish a separately selected balance-of-payments CA series. RWNEOWQ027S is ROW net-worth revaluation (US total VA is its negative); equity FU components are transactions. AHP retains statistical discrepancies and other-volume changes. ROW net worth; ROW net lending.

F5. Equity revaluation and price/FX separation

Live scope discrepancy; current bridge unresolved.

BOGZ1FR263181105Q, BOGZ1FR263081005Q and BOGZ1FR263081115Q are quarterly revaluations in million USD, not annualized transaction flows. BEA Concepts and Methods separates transactions, price/FX revaluation and other changes. Matching FR/FU/stock universes is required.

Broad liability FR; Legacy liability FR; BEA 2026 methods, chs. 8/24.

F6. Safe yields and Treasury holdings

Live series; Treasury coverage notes checked.

TB3MS: monthly average 3-month bill rate, percent at an annual discount-basis quote; saved through Aug 2026. ROWTSEQ027S: quarterly foreign Treasury holdings. TIC Table 5 includes multiple maturities and custody-based country attribution; it is not a model wealth denominator.

TB3MS; TIC Table 5.

F7. Capitalization, operating-asset value and distributions

Live Fed alternatives; local AHP construction retained.

AHP operating-asset values: LM102010405Q+LM792010405Q. FCF: (FU106402101+FU796402101)-(FU106220001+FU796220001)-(FU105050985+FU795015085), with BOGZ1 prefix and Q suffix. These FU inputs are quarterly million-USD flows, not SAAR. Gross-dividend alternatives FU106121001A+FU796121001A need consolidation. AHP justifies EV/FCF using financing neutrality; V6 additionally needs an entry/issuance cash-flow bridge. Operating-asset value; NFC dividends.

F8. Dividends received on foreign equity

Exact official CSV probe, 23 September 2026 local time.

BEA NIPA Table 4.1 line 11; account B3375C; FRED B3375C1Q027SBEA. Quarterly billions of USD at SAAR: multiply by 1000/4 for a quarterly million-USD payout. The sample, ROWEISQ027S and FR263181105 yield 10 observations, 2024Q1–2026Q2. Excludes separately reported reinvested earnings. AHP App. A4 uses the monetary-payout yield; tax-repatriation spikes and instrument/currency coverage need explicit treatment. OpenEcon timed out; exact-source CSV succeeded.

Exact dividend series; NIPA Table 4.1; BEA dividend concepts.

M1. Model, prior crosswalk and workflow

Local source review; no baseline changes or new estimation.

V6_calibration_core.tex; V6 manuscript; TwoCountryProductionOLG.jl; annual empirical mapping registry and method notes; FinAI AGENTS.md, SMU_RESEARCH and SMU_PROVIDERS. Source authority, access success and economic mapping are separate assessments.